

34. Evaluate $\int_{(0,0,0)}^{(1,2,3)} 2xydx + (x^2 - z^2)dy - 2yzdz$.

35. Show that $\vec{f} = (x + 2y + 4z)\vec{i} + (2z - 3y - z)\vec{j} + (4x - y + 2z)\vec{k}$ is conservative and find its scalar potential.

(5 × 9 = 45 marks)

Section D

*Answer any **one** question.*

The question carries 15 marks.

36. Verify Gauss's divergence theorem for $\vec{f} = x\vec{i} + y\vec{j} + z\vec{k}$ over the sphere of radius a centred at the origin.

37. (a) Evaluate $\int_{(1,0,0)}^{(0,1,0)} \sin y \cos x dx + \cos y \sin x dy + dz$.

(b) If S is a closed surface enclosing a volume V , then prove that $\iint_S (\text{curl}) \vec{f} \cdot n dS = 0$.

38. Using Green's theorem find the counter clockwise circulation and outward flux for the field $\vec{f} = (x^2 + 4y)\vec{i} + (x + y^2)\vec{j}$ and the square C bounded by $x = 0, x = 1, y = 0, y = 1$.

(1 × 15 = 15 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Mathematics

MTS 5D 04—MATHEMATICS FOR DECISION-MAKING

(2019 Admissions)

Time : Two Hours

Maximum : 60 Marks

Section A*Answer at least eight questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 24.*

1. Define Statistics.
2. Distinguish between population and sample.
3. What is meant by frequency distribution ?
4. Define Median.
5. Write down the sample space when a six sided die is rolled.
6. Identify the type of probability in the statement : The probability that you will get an A on your next test is 0.9.
7. Define mutually exclusive events.
8. State the addition rule of probability of two events.
9. What is a discrete random variable ?
10. Find the missing probability from the following probability distribution :

x	:	0	1	2	3	4
$P(x)$	:	0.07	0.20	0.38	?	0.13
11. Find the mean and variance of a binomial distribution with $n = 50$ and $p = 0.4$.
12. In a geometric distribution, what does the random variable measure ?

(8 × 3 = 24 marks)

Turn over

Section B

Answer at least five questions.

Each question carries 5 marks.

All questions can be attended.

Overall Ceiling 25.

13. Explain the different levels of measurement of data.
14. What is the difference between stratified sample and systematic sample ?
15. Give the Classical and Empirical definitions of probability.
16. A coin is tossed and then a die is rolled. Find the probability of tossing a head and then rolling a 6.
17. A corporation has six male senior executives and four female senior executives. Four senior executives are chosen at random to attend a technology seminar. What is the probability of choosing two male and two female senior executives ?
18. The number of hours students in a college class slept the previous night is given below :

Hour	:	4	5	6	7	8	9	10
Students		1	6	8	20	9	4	2

Construct a probability distribution.

19. A particular surgery has 90 % chance of success. The surgery is performed on three patients. Find the probability of the surgery being successful on exactly two patients.

(5 × 5 = 25 marks)

Section C

Answer any one question.

The question carries 11 marks.

20. The marks of students in two subjects A and B are given below. Find the co-efficient of variation and compare the results.

A		35	70	45	62	55	72	90	44	62	75
B	:	88	73	55	82	48	95	82	95	48	84

21. About 60 % cancer survivors are ages 65 years and older. Six cancer survivors are selected at random ask them whether they are 65 years of age and older. Construct a probability distribution for the number of cancer survivors of this age group. Also find the mean, variance and standard deviation.

(1 × 11 = 11 marks)

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021**(CBCSS—UG)****Mathematics****MTS 5D 03—LINEAR MATHEMATICAL MODELS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 15****Maximum : 15 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 15.
2. The candidate should check that the question paper supplied to him/her contains all the 15 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 5D 03—LINEAR MATHEMATICAL MODELS

(Multiple Choice Questions for SDE Candidates)

1. What is the slope of line passing through the points (0,1) and (1,1) ?
(A) 1. (B) -1.
(C) 0. (D) 0.5.
2. Equation of a line whose x -intercept is 1 and y -intercept is -1.
(A) $x + y = 1$. (B) $x - y = 1$.
(C) $x + y = -1$. (D) $x - y = -1$.
3. P : Slope of a horizontal line is 1.
Q : Slope of a vertical line is undefined.
(A) P is true and Q is false. (B) P is false and Q is true.
(C) P and Q are false. (D) P and Q are true.
4. Find the equation of the line that passes through the point (3, 5) and is parallel to the line $2x + 5y = 4$.
(A) $5x + 2y = 25$. (B) $2x - 5y = -19$.
(C) $2x + 5y = 31$. (D) $2x + 5y = -19$.
5. Find the equation of the line that passes through the point (0,1) and is perpendicular to the line $x = 0$.
(A) $y = 0$. (B) $y = 1$.
(C) $y = -1$. (D) None of the above.
6. What is the solution of the system of linear equation given below ?
 $x + y = 1$
 $x - y = -1$
(A) (0, 1). (B) (0, -1).
(C) (1, 0). (D) (-1, 0).

7. $x - 3y = 1$ the linear system of equation has :
 $x - 3y = -1$

(A) Unique solution. (B) Infinitely many solutions.
(C) No solution. (D) None of the above.

8. $2x - 6y = 4$
 $3x - 9y = 6$ the linear system of equation has :

(A) Unique solution. (B) Infinitely many solutions.
(C) No solution. (D) None of the above.

9. $\begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix} + \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix} = \underline{\hspace{2cm}}.$

(A) $\begin{bmatrix} -1 & 10 \\ 16 & 6 \end{bmatrix}.$

(B) $\begin{bmatrix} 1 & 0 \\ 16 & 6 \end{bmatrix}.$

(C) $\begin{bmatrix} -1 & 12 \\ 16 & -6 \end{bmatrix}.$

(D) $\begin{bmatrix} -1 & 0 \\ 16 & -6 \end{bmatrix}.$

10. Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$. Then A B is defined if and only if :

(A) $m = p$. (B) $m = q$.
(C) $n = p$. (D) $n = q$.

11. $A = \begin{bmatrix} 5 & -6 \\ 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 6 \\ 8 & -3 \end{bmatrix}$. Then $AB = \underline{\hspace{2cm}}.$

(A) $\begin{bmatrix} -68 & 48 \\ 40 & 19 \end{bmatrix}.$

(B) $\begin{bmatrix} -68 & 48 \\ -104 & 21 \end{bmatrix}.$

(C) $\begin{bmatrix} -68 & 42 \\ 40 & 21 \end{bmatrix}.$

(D) $\begin{bmatrix} -68 & 48 \\ 40 & 21 \end{bmatrix}.$